Digital Technology Use and Perceived Community Identification Among Young Adults

Alonso Flores-Sáez

University of California, Irvine
Demographic and Social Analysis (DASA)
Dr. Edward Telles

June 15th, 2023

Acknowledgements

I would like to extend my gratitude to the School of Social Sciences at the University of California, Irvine for the opportunity to develop my research interests in the process of pursuing the Demographic and Social Analysis MA program and for the support in developing the skills and methodological understanding to do so. I would also like to thank my advisor, Dr. Edward Telles, as well as director Dr. Rachel Goldberg, for their sound and measured guidance. Most importantly, I would like to thank the Department of Sociology as a community, from the faculty to the graduate student body, for fostering a deep sense of wonder in the pursuit of knowledge. The breadth of wisdom that I encountered here has been nothing short of awe-inspiring, and I am content to end my time within this community with the confidence that it has been and continues to be explored with enduring creativity by wonderfully capable people. I will remember my time here fondly, and I will keep the knowledge and goodwill that was shared with me close as I continue on my journey. Thank you. Truly.

Abstract

Recent attention has been placed by scholars and public officials on the rise of loneliness in the United States and its potential association with digital technology use, particularly with social media. An advisory on the "epidemic of loneliness" by the U.S. Surgeon General indicates that research has found mixed results regarding the association between increased time spent on social media and perceived loneliness. The intent of this project is to measure such an association in young adults, who are likely to be the first cohort in the U.S. to have spent formative adolescent years using social media. Using data from the latest available wave of the Panel Study of Income Dynamics' (PSID) Transition to Adulthood Supplement (TAS), I used ordered logistic regression to determine the direction and statistical significance of the association between respondents' frequency of digital communication technology use and reported frequency of identification with an established community. The results find that general communication technology use frequency is associated with a statistically significant increase in sense of community, net of psychological, demographic, and social participation controls. The partial model also finds that respondents' sense of community is 40% more likely to decrease every year between the ages of 17 and 28 until community participation measures are introduced, which find that college enrollment and respondents' race present some of the strongest factors in favor of and against community identification, respectively. This points to a more structural culprit to the epidemic of loneliness.

Introduction

Digital Communication Technology

The ubiquity of digital communication technology is readily evident in our daily life. In 2019, the 50% threshold for global internet penetration was reached, meaning that a majority of human beings on earth now have access to the internet by some means. In the case of the United States, the percentage of people who use the internet stands at 92% as of 2021.[1] This global and national change has fundamentally changed the means and logics of information, the repercussions of which can be found in nearly all facets of contemporary social life.

Many if not most of our daily processes have become reliant on some form of digital mediation, from online banking to remote office work to digitized home appliances. The result of this is that, in most cases, our social processes have come to adapt to the structural frameworks of the media technologies that have come to house them.[2] Most interestingly, digital information technology has even succeeded in translating the spatial elements of social interaction into mediated formats. Even before a global pandemic generated an explosive demand for such features, many children and young people often found themselves communicating within virtual worlds via personalized avatars, and U.S. adults over 30 overwhelmingly reported social media use on platforms like Facebook.[3] As a result, new etiquettes and use cultures have evolved and adapted around the constraints of such technologies.

---

[1] World Bank, " Individuals using the Internet (% of population)," The World Bank Group. Accessed March 13, 2023, https://data.worldbank.org/indicator/IT.NET.USER.ZS

[2] Stig Hjarvard, *The Mediatization of Culture and Society* (New York: Routledge, 2013).

[3] "Social Media Fact Sheet." Pew Research Center. Accessed March 13, 2023. https://www.pewresearch.org/internet/fact-sheet/social-media/#panel-3814afe3-3f3c-4623-910b-8a6a37885ab8

Loneliness and Community

At the same time that digital media has come to dominate public life, there has been a drastic rise in levels of isolation in the United States. The U.S. Surgeon General issued an advisory on "Our Epidemic of Loneliness and Isolation" in May of this year, which described the increase of social isolation (time spent alone) and perceived loneliness, as well as its implication on public health.[4]

Among the many factors of weakened social ties described by the Surgeon General, the role of social media is discussed. Some studies have found that increased social media is associated with higher rates of perceived social isolation.[5] Others have found that image-based social media use is associated with increased happiness and decreased loneliness, possibly due to a higher level of social presence than text-based social networking platforms.[6] Certain scholars have taken a middle ground, arguing that social media presents a balance of new risks and opportunities for fostering social connections and intimacy, allowing people to overcome barriers to entry into social space through technological aids.[7] It is, however, widely agreed upon that there must be some association between the two.

The intention of this project is to explore the association between digital communication technology use and reported sense of community. Perceived community identification, while similar to perceived social isolation or loneliness, captures more of a

---

[4] United States. Public Health Service. Office of the Surgeon General. *Our Epidemic of Loneliness and Isolation: The U.S. Surgeon General's Advisory on the Healing Effects of Social Connection and Community*. Washington, DC: U.S. Department of Health and Human Services, Public Health Service, Office of the Surgeon General, 2023.

[5] Brian A. Primack et al., "Social Media Use and Perceived Social Isolation Among Young Adults in the U.S.," *American Journal of Preventive Medicine* 53, no. 1 (July 1, 2017): 1–8, https://doi.org/10.1016/j.amepre.2017.01.010.

[6] Matthew Pittman and Brandon Reich, "Social Media and Loneliness: Why an Instagram Picture May Be Worth More than a Thousand Twitter Words," *Computers in Human Behavior* 62 (September 1, 2016): 155–67, https://doi.org/10.1016/j.chb.2016.03.084.

[7] Livingstone, Sonia. "Taking Risky Opportunities in Youthful Content Creation: Teenagers' Use of Social Networking Sites for Intimacy, Privacy and Self-Expression." New Media and Society 10, no. 3 (2008): 393–411. https://doi.org/10.1177/1461444808089415.

sense of an individual's place within formalized social environments. This distinction allows for a less agentic approach to social connection, placing more emphasis on perceptions of social structure and position rather than individual experience.

Research Questions and Hypotheses

The debate I have laid out in the previous section relies on a paradox between two hypotheses of media use effects on social behavior that I believe can apply to this given case. The first hypothesis is the "rich get richer" hypothesis, wherein people with greater social aptitude see increased benefit from their internet use and more introverted people will see worse outcomes.[8] The second hypothesis is the social compensation hypothesis, wherein socially anxious subjects will ultimately prefer digital communication technology to face-to-face interaction and see a benefit in technological assistance in social life.[9]

My hypothesis is that, contrary to certain findings, increased frequency of digital communication technology use will be associated with an increase in frequency of reported sense of community within the sample. I also hypothesize that technology use will have a positive interaction effect on other factors of community identification, such as race and gender, helping people of more marginalized identities reduce the existing inequalities in access to community, as is consistent with the social compensation hypothesis.

---

[8] Robert Kraut et al., "Internet Paradox Revisited," *Journal of Social Issues* 58, no. 1 (January 1, 2002): 49–74, https://doi.org/10.1111/1540-4560.00248.

[9] Patti M. Valkenburg and Jochen Peter, "Online Communication Among Adolescents: An Integrated Model of Its Attraction, Opportunities, and Risks," *Journal of Adolescent Health* 48, no. 2 (February 1, 2011): 121–27, https://doi.org/10.1016/j.jadohealth.2010.08.020.

Methods

Sampling Data

Due to the time constraints of the DASA program, existing survey data was needed in order to conduct an analysis. My chosen survey was the University of Michigan's Panel Study of Income Dynamics (PSID), particularly its Transition to Adulthood Supplement (TAS), which seeks to measure the period between childhood and financial independence during young adulthood. The PSID is a panel survey of U.S. households and began in 1968 as a study on income and poverty dynamics and has expanded to include a number of health, employment, and lifestyle variables.[10]

Beginning in 1997, the PSID began to conduct the Childhood Development Supplement (CDS), in which up to two children under the age of 12 per PSID household were selected in order to capture early stages of development and social capital formation. This cohort was then re-interviewed in up to two additional waves in 2002 and 2007. However, by the time of the third (2007) wave, the oldest members of the cohort had entered adulthood and were no longer eligible. These respondents were included in the very first biennial TAS wave beginning in 2005. In 2017, the TAS was redesigned, expanding eligibility to all children of PSID households rather than a sample of up to two children.[11] The 2017 wave also saw the introduction of survey questions about technology use habits that are of interest for my analysis. Given that there were only two biennial waves of TAS data that included these variables of interest at the time of selecting a dataset, I opted to go with the latest public release TAS wave, which was conducted in 2019. The 2019 TAS wave (TAS19) consists of 2,595 respondents between the ages of 17 and 28.

---

[10] PSID Main Interview User Manual: Release 2021. Institute for Social Research, University of Michigan, March, 2021.

[11] "Panel Study of Income Dynamics, Transition into Adulthood Supplement 2019: User Guide," Institute for Social Research, University of Michigan, 2021.

Measures

The first task in manipulating the data was identifying and cleaning my dependent variable. The TAS19 variable for community identification asks the respondent to answer the following question:

"In the last month, how often did you feel... That you belonged to a community like a social group, your school, or your neighborhood? Would you say never, once or twice, about once a week, 2 or 3 times a week, almost every day, or every day?"[12]

Answers ranging from never to every day were originally coded from 1 to 6, and missing answers were coded as 9. I recoded this scale to run from 0 to 5 instead, and dropped the missing cases (of which there were 9). I did this primarily because I was Initially using OLS regression in my analysis due to a lack of familiarity with ordered logit, and I felt that this coding would make for a more intuitive scale. Once I began working with ordered logistic regression, I kept the scale, as I still felt that it was more intuitive to code "never" as 0 than 1. Tabulation and summary statistics of this variable show that the largest group of respondents (27.92%) reported never feeling a sense of community (see table 1 in index).

Next, I had to repeat this process for my independent variable of choice, in this case frequency of digital communication use. The TAS asks the following question regarding the frequency of five types of technology use:

In the past 30 days how often have you…

a) used a computer or other electronic device (such as a tablet, smartphone or gaming console) to play games?
b) Sent or received email?
c) Sent or received text messages?
d) Interacted with friends or family on a social media site (like Facebook, Instagram, or Twitter)?
e) Followed topics or people that interest you on websites, blogs, or social media sites (like Facebook, Instagram, or Twitter), not including following or interacting with friends or family online?

[12] From the TAS 2019 Codebook

(Would you say every day, a few times a week, once a week, less than once a week or never?)[13]

These responses were coded in reverse, with 1 representing "every day" and 5 representing "never" while 8 represented "I don't know" and 9 denoted missing data. In keeping with my previous intuition, this coding was reversed, ranging from 0 "never" to 4 "every day". Missing data and nonresponses were both dropped. Afterwards, I coded a separate variable to equal the mean of all five responses as an aggregate scale of technology use. Tabulation and summary statistics of this technology use variable and its constituent variables shows that a majority of respondents reported daily engagement with every form of technology stated (see table 2 in index).

Covariates/Controls

In order to isolate the association between technology use frequency and community identification and identify potential confounding variables, I looked to determine covariates for my regression model. Of available variables in the TAS19 codebook, I identified three general categories of potential influences: psychological effects, demographic effects, and social participation effects. These will all be included in my full ordered logistic regression model.

Psychological Factors

An early concern in this analysis was the difficulty of determining causality with cross-sectional data. The "rich get richer" hypothesis has received more empirical support than its counterpart, which could suggest that it is actually people who are already more socially apt that are more likely to use these technologies frequently, and thus see increased

[13] Ibid.

benefit.[14] As such, I chose to include a number of psychological and perception variables into my regression model in order to rule out the possibility of the confounding influence of internal perceptions.

The first of such measures are psychological and emotional well-being. Using the same frequency scale as the community identification variable, respondents were asked the following:

In the last month, how often did you feel…

j) Good at managing the responsibilities of your daily life?
k) That you have warm and trusting relationships with other people?
l) That you have experiences that challenged you to grow or become a better person?
m) Confident to think or express your own ideas and opinions?
n) That you liked your personality?
o) That your life had a direction or purpose?

The same cleaning process was applied to these six variables. The TAS provides an aggregate "psychological well-being" subscale as a separate variable, taking the average of the six variables for each respondent. However, this subscale rounds the average value to the nearest integer, so I generated an alternate scale based on a true average for accuracy. The same goes for the "emotional well-being" subscale, which measures the average response to the following questions:

In the last month, how often did you feel…

a) Happy?
b) Interested in life?
c) Satisfied?

In addition to psychological well-being, self-esteem was also measured as a covariate in the model. The TAS utilizes the Rosenburg Self-Esteem Scale, which asks respondents the following ten questions:

Please tell me your level of agreement with the following statements

a) I feel that I'm a person of worth, at least on an equal basis with others.

[14] Valkenburg and Peter, 'Online Communication Among Adolescents

b) I feel that I have a number of good qualities.
c) All in all, I am inclined to feel that I am a failure.
d) I am able to do things as well as most other people.
e) I feel I do not have much to be proud of.
f) I take a positive attitude toward myself.
g) On the whole, I am satisfied with myself.
h) I wish I could have more respect for myself.
i) I certainly feel useless at times.
j) At times I think I am no good at all.

Respondents are asked to record their level of agreement using a four-point likert scale, ranging from strongly disagree to strongly agree. Answers are then assigned values from 0 to 3 according to how positive the answer is (you'll notice that half of the questions describe negative attitudes and half of them describe positive attitudes) and then the total sum across the ten variables comprises the final self-esteem scale. I was satisfied with this coding process, so the self-esteem variable was left intact.

Demographic Factors

Like most facets of health and well-being, the benefits of community and the effects of social isolation are subject to long-standing demographic inequalities.[15] As such, demographic covariates were included in my regression model, including respondents' race, gender, age, and income.

Black respondents are heavily oversampled (42.8% of the sample population) and White, Hispanic, and Asian respondents are undersampled (42.35%, 11.69%, and 2.02%, respectively). At the time of my analysis, the PSID had not yet released cross-sectional weights for TAS19, so the 2019 PSID cross-sectional weights were used in their place, as per the recommendation of the TAS User Guide.[16] I used the STATA *pweight* command in order to do this. Due to the nature of the TAS, 49.25% of the sample population reported earning

[15] Viji Diane Kannan and Peter J. Veazie, "US Trends in Social Isolation, Social Engagement, and Companionship – Nationally and by Age, Sex, Race/Ethnicity, Family Income, and Work Hours, 2003–2020," *SSM - Population Health* 21 (March 1, 2023): 101331, https://doi.org/10.1016/j.ssmph.2022.101331.

[16] Transition into Adulthood Supplement 2019: User Guide

less than $10,000 in the previous year, so it is to be expected that the effects of income on this regression may not be representative of the broader population. Gender identity variables are sealed from public use by the PSID, resulting in a very reductive variable which purely describes sex, which we will use as a proxy for measuring gender (though I cannot know how many trans and gender non–conforming respondents there are in the sample, I suspect that the number is not high). Both age and age^2 are included in the model, in order to control for a non-linear association between community and age over time.

Social Participation Factors

It is worth noting that, in the survey question for the community identification variable, specific examples of community are mentioned, two of the three of which are explicitly geographically bounded. It is well established that while perceptions of things like social isolation and community are dependent on mental states, these perceptions are still primarily determined by our physical surroundings.[17] With this in mind, I looked to include all available survey variables which have to do with social participation. Section A of the TAS19 survey focuses "on involvement over the last 12 months in the community including volunteering and community service, group organizations, and sports participation, as well as the type of organization and frequency of participation."[18] My intention was to include all of the questions from section A that constitute dichotomous variables, which allowed me to control for involvement in the arts, organized sports, clubs, and volunteering.

One of the well-documented facets of the decline in social connection over the last 20 years is the decline in religious participation, which dropped from 70% to 45% from 1999 to

[17] Lisa F. Berkman et al., "From Social Integration to Health: Durkheim in the New Millennium.," *Social Science & Medicine* 51 (2000): 843–57, https://doi.org/10.1016/S0277-9536(00)00065-4.

[18] Transition into Adulthood Supplement 2019: User Guide

2020.[19] This has been and remains one of the strongest institutions of public life in the United States, and so I also included controls respondents' frequency of religious participation, which was recoded on a scale from 0 to 5, capturing answers of "Never" "Yearly" "Monthly" "Biweekly" "Weekly" and "Monthly."

Another of the strongest institutions which foster community is education. I included two measures for this factor in my model: a dichotomous variable indicating whether the respondent had attended a higher learning institution of any kind, and a variable which asked respondents what percentage of their friends had attended a higher learning institution of some kind, scaled from 0 to 4, in increments of 25%.

Findings

I developed several partial models of my regression in order to better demonstrate the significance of my independent variable with the addition of added covariates. The results are shown below, represented in odds ratios.

[19] Jones JM. US church membership falls below majority for first time. Gallup News. 2021.

Ordered Logistic Regression Analysis of Community Identification Indicators

	Dependent variable: Community Identification Frequency							
	Model 1		Model 2		Model 3		Model 4	
Technology use frequency	1.650***	(0.129)	1.518***	(0.125)	1.444***	(0.125)	1.242***	(0.116)
Psychological well-being			1.647***	(0.126)	1.657***	(0.133)	1.649***	(0.136)
Emotional well-being			1.217***	(0.086)	1.227***	(0.041)	1.300***	(0.100)
Rosenburg self-esteem scale			1.051***	(0.013)	1.071***	(0.014)	1.062***	(0.015)
Race (Ref: White)								
Hispanic, Latino, or Spanish					0.358***	(0.059)	.396***	(0.066)
Black or African American					0.478***	(0.072)	.557***	(0.084)
Asian					1.012	(0.230)	.831	(0.227)
American Indian or Alaskan Native					0.811	(0.692)	1.373	(1.509)
Middle Eastern or North African					0.182*	(0.155)	.371	(0.394)
Native Hawaiian or Other Pacific Islander					3.383***	(0.644)	3.568***	(0.762)
Some other race, ethnicity, or origin					1.441	(0.702)	1.079	(0.437)
Gender (Ref: Male)								
Female					1.048	(0.110)	0.967	(0.107)
Other					2.045	(0.991)	2.673*	(1.603)
Earnings in previous year					0.998	(0.003)	0.998	(0.003)
Age					0.588**	(0.139)	0.681	(0.178)
Age²					1.011**	(0.005)	1.009	(0.005)
Social Participation								
Arts							1.535***	(0.244)
Sports							1.512***	(0.227)
School clubs							1.515***	(0.023)
Social action groups							1.414*	(0.311)
Volunteering							1.426***	(0.179)
Frequency of religious attendance							1.139***	(0.012)
Higher educ. enrollment							2.157***	(0.328)
Friends' higher educ. Enrollment							1.140***	(0.053)
Cut Points								
1	0.439	(0.262)	3.459	(0.349)	-2.916	(2.603)	-0.139	(2.853)
2	1.037	(0.263)	4.121	(0.351)	-2.225	(2.607)	0.660	(2.856)
3	1.426	(0.264)	4.572	(0.356)	-1.748	(2.608)	1.213	(2.856)
4	1.999	(0.266)	5.278	(0.364)	-1.017	(2.611)	2.011	(2.859)
5	3.005	(0.270)	6.473	(0.380)	0.226	(2.614)	3.336	(2.861)
N	2569		2494		2382		2240	
Pseudo R^2	0.0101		0.0734		0.0931		0.1318	
p-value	0		0		0		0	

**** p<.01, ** p<.05, * p<.1*

Technology Use Frequency

As we can see, the aggregate technology use variable maintains a statistically significant association with community identification throughout the four models, net of all covariates ($p<0.01$). This points to the possibility that increased technology use frequency has a significant positive effect on the likelihood of reporting a more frequent sense of community by as much as 27%. In the simplest partial model with no statistical controls, the pseudo-R^2 of the logistic regression suggests that the model only explains about 1% of the variation, but the fact that technology use frequency remains consistent with the introduction of more significant predictors of community is noteworthy. When the model is expanded to include all technology use measures separately, we see that the only significant positive technology use indicator is following interests and people on social media (see table 3 in index). This could lend credence to the social compensation hypothesis, as it suggests that the activity which is most associated with increased likelihood of community identification frequency is following and engaging with people and interests outside of one's given surroundings.

Demographic Factors

The full ordered logistic regression model yields some very interesting results in terms of demographic factors in community identification. Black and Hispanic respondents show a significantly lower likelihood of reporting more frequent feelings of belonging to community than their white counterparts. This contradicts certain social isolation studies, which have shown much higher levels of social isolation in Black respondents but much lower levels of social isolation in Hispanic subjects.[20] Nevertheless, it seems that the Hispanic respondents along with Black respondents are less likely than white respondents to

[20] Kannan and Veazie, *US Trends in Social Isolation, Social Engagement, and Companionship*.

frequently identify with a set community by as much as 61% and 45%, respectively. The reasons for this are likely tied to existing barriers to access to public space, institutions, and housing for racial minorities. The results for Hawaiian and other Pacific Islanders in the sample raises an interesting question, however. While racial inequalities exist for all groups mentioned, it is also true that racial categories can serve to form strongly bounded social networks and rigidly defined communities, as is the case with the Hawaiian population, who show a 357% greater likelihood of increase frequency of community identification than white respondents. It also raises the distinction between native and non-native minorities, where the preservation of shared histories and culture becomes crucial and is not always accessible to pan-ethnic groups or diasporic populations such as Black and Hispanic Americans.

It seems that there are no significant associations between community identification and gender, which is surprising given the well-established inequalities that fall along these lines. Additionally, wage/income showed no statistical significance within the model, which I had predicted from the outset due to the nature of the study. Lastly, age and age^2 were both significant at the 95% level of confidence until social participation measures were introduced to the model. I suspect that this is because access to the social participation measures laid out by the TAS are very reliant on institutional backing that people lose access to as they age. It is much less likely that someone will participate in an organized sports team or a school club at 28 than at 17, confounding the influence of age and age^2 on community identification. I found the suggestion that community identification frequency would decrease with age, with the effect subtly reducing over time to be quite compelling. However, with the introduction of these social participation measures, I can no longer be confident that this is the case.

Social Participation Factors

It is clear from the model that social participation is the strongest indicator of frequency of feeling belonging to a community. This is not particularly revelatory, as the act

of engaging with a community and belonging to it are very similar and closely related. It is worth noting, however, the outsized association between higher education enrollment and increased community identification frequency, as respondents who have at any point attended college are 215% more likely to report increased community id frequency than those who have not. This effect is much stronger than that of religious frequency, which at its highest could result in a 56% higher likelihood of this outcome.

Discussion & Conclusions

The role of digital communication technology in community identification seems to be a positive one, though its effects are minimal in the scope of all of the influences on people's sense of community. As can be seen from the regression model, demographic and social participation factors show much stronger associations, demonstrating inequalities that are readily visible in everyday life. The widening gap between the college educated and non-college educated people in the United States is also demonstrated here. As social isolation and lack of community pose increased mortality risks, the educational gap in the United States is also one of differential health outcomes.[21] As such, barriers to access to institutionally backed community seems to be a significant driver of the divide in community identification in the United States.

Though technology use measures showed a significant positive association with community identification, there were no significant interaction terms that could suggest an improvement in other inequalities with their use. As such, I cannot come to a strong conclusion on either the rich-get-richer hypothesis or the social compensation hypothesis. I

[21] Julianne Holt-Lunstad et al., "Loneliness and Social Isolation as Risk Factors for Mortality: A Meta-Analytic Review," *Perspectives on Psychological Science* 10, no. 2 (March 1, 2015): 227–37, https://doi.org/10.1177/1745691614568352.

can, however, speculate on the role of digital communications in the future of community identification.

Much of the attention on social isolation and the decrease in social connectivity has to do with concerns over the dissolution of place in modern society, as globalization and urbanization continue to take shape both nationally and globally.[22] As space continues to be produced and commodified and city boundaries begin to collide with one another in the process of urbanization and economic development, there is a sense of belonging that is lost, resulting in weaker social ties.[23] However, it is also the case that urbanization results in a much more dense network of informal ties, as people begin to rely on each other for more things, resulting in more specific forms of community formation and new subcultural associations.[24] It can be argued that the same process is occurring with digital communication structures. As communication infrastructures continue to develop and people are placed in closer contact with one another via digital networking, it is possible that there will be a reconstitution of place according to these new infrastructural considerations.

Unfortunately, there are a number of factors that limited the effectiveness of this study. The first and most pressing constraint was time, which forced me to select a dataset for analysis quickly and limited me to publicly accessible data. This also meant that I was not able to make alterations to the design of the study according to my research questions, leaving me relying on fairly simple and broad measures from which to conduct my analysis. In the future, I would like to conduct a more personalized survey of young adults and specifically sample those who spend very large amounts of time socially interacting on the internet. A study of the self-perceptions and community identifications of those people who

[22] Fischer, C S, and R K Merton. *The Urban Experience*. Harcourt Brace Jovanovich, 1976.

[23] Gotham, Kevin Fox, and Arianna J King. "Urbanization." In *The Wiley Blackwell Companion to Sociology*, 267–82, 2019. https://doi.org/https://doi.org/10.1002/9781119429333.ch16.

[24] Fischer and Merton, *The Urban Experience*

spend the majority of their time on the internet could yield new and insightful results towards an understanding of the future of mediated socialization.

Bibliography

World Bank, " Individuals using the Internet (% of population)," The World Bank Group. Accessed March 13, 2023, https://data.worldbank.org/indicator/IT.NET.USER.ZS

Hjarvard, Stig. "The Mediatization of Society A Theory of the Media as Agents of Social and Cultural Change." *Nordicom Review* 29 (January 1, 2008).

"Social Media Fact Sheet." Pew Research Center. Accessed March 13, 2023. https://www.pewresearch.org/internet/fact-sheet/social-media/#panel-3814afe3-3f3c-4623-910b-8a6a37885ab8

United States. Public Health Service. Office of the Surgeon General. *Our Epidemic of Loneliness and Isolation: The U.S. Surgeon General's Advisory on the Healing Effects of Social Connection and Community*. Washington, DC: U.S. Department of Health and Human Services, Public Health Service, Office of the Surgeon General, 2023.

Brian A. Primack et al., "Social Media Use and Perceived Social Isolation Among Young Adults in the U.S.," *American Journal of Preventive Medicine* 53, no. 1 (July 1, 2017): 1–8, https://doi.org/10.1016/j.amepre.2017.01.010.

Matthew Pittman and Brandon Reich, "Social Media and Loneliness: Why an Instagram Picture May Be Worth More than a Thousand Twitter Words," *Computers in Human Behavior* 62 (September 1, 2016): 155–67, https://doi.org/10.1016/j.chb.2016.03.084.

Livingstone, Sonia. "Taking Risky Opportunities in Youthful Content Creation: Teenagers' Use of Social Networking Sites for Intimacy, Privacy and Self-Expression." New Media and Society 10, no. 3 (2008): 393–411. https://doi.org/10.1177/1461444808089415.

Robert Kraut et al., "Internet Paradox Revisited," *Journal of Social Issues* 58, no. 1 (January 1, 2002): 49–74, https://doi.org/10.1111/1540-4560.00248.

Patti M. Valkenburg and Jochen Peter, "Online Communication Among Adolescents: An Integrated Model of Its Attraction, Opportunities, and Risks," *Journal of Adolescent Health* 48, no. 2 (February 1, 2011): 121–27, https://doi.org/10.1016/j.jadohealth.2010.08.020.

PSID Main Interview User Manual: Release 2021. Institute for Social Research, University of Michigan, March, 2021.

"Panel Study of Income Dynamics, Transition into Adulthood Supplement 2019: User Guide," Institute for Social Research, University of Michigan, 2021.

Viji Diane Kannan and Peter J. Veazie, "US Trends in Social Isolation, Social Engagement, and Companionship – Nationally and by Age, Sex, Race/Ethnicity, Family Income, and Work

Hours, 2003–2020," *SSM - Population Health* 21 (March 1, 2023): 101331, https://doi.org/10.1016/j.ssmph.2022.101331.

Lisa F. Berkman et al., "From Social Integration to Health: Durkheim in the New Millennium.," *Social Science & Medicine* 51 (2000): 843–57, https://doi.org/10.1016/S0277-9536(00)00065-4.

Jones JM. US church membership falls below majority for first time. Gallup News. 2021.

Julianne Holt-Lunstad et al., "Loneliness and Social Isolation as Risk Factors for Mortality: A Meta-Analytic Review," *Perspectives on Psychological Science* 10, no. 2 (March 1, 2015): 227–37, https://doi.org/10.1177/1745691614568352.

Fischer, C S, and R K Merton. *The Urban Experience*. Harcourt Brace Jovanovich, 1976.

Gotham, Kevin Fox, and Arianna J King. "Urbanization." In *The Wiley Blackwell Companion to Sociology*, 267–82, 2019. https://doi.org/https://doi.org/10.1002/9781119429333.ch16.

Index

Table 1: Tabulation of Community Identification Frequency

	Freq.	Percent	Cum.
Never	722	27.92	27.92
once or twice	301	11.64	39.56
About once a week	237	9.16	48.72
Two or three times a week	303	11.72	60.44
Almost every day	475	18.37	78.81
Every day	548	21.19	100.0
Total	2586	100.00	

Table 2: Summary of technology use frequency

Frequency	Gaming	Email	Text	Interact	Follow
Never	322	97	26	139	225
Less than once a week	318	210	23	137	250
Once a week	202	189	46	122	267
A few times a week	572	568	213	586	709
Every day	1178	1524	2279	1603	1140
Total	2592	2588	2587	2587	2591

Table 3: Ordered Logistic Regression (Model 4 Expanded)

Community ID	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
Gaming	.983	.038	-0.46	.648	.911	1.059	
Email	1.097	.072	1.42	.155	.966	1.247	
Texting	.75	.088	-2.45	.014	.596	.944	**
Interacting	1.048	.068	0.72	.472	.922	1.191	
Following	1.173	.067	2.78	.005	1.048	1.313	***